

Alex Lee

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EDUCATION

University of New South Wales

Bachelor of Computer Science

Sydney, Australia

May 2022 – Aug 2026

- WAM: 81 (2025 Academic Year WAM: 87)
- COMP3311 - 91 HD, COMP3231 - 89 HD, COMP1511 - 88 HD, COMP3900 - 85 HD
- UniHack 2025: Best Design Award & Social Impact Award (Only 10 / 135 participating teams awarded)
- Computer Science Mentor for Colombo House (2024), Trainee Lead for UNSW Development Society (2025)

EXPERIENCE

Software Engineer Intern

Google

Nov 2025 – Present

Sydney, Australia

- Google Maps SRE, Geo-Data Team
- Engineered high-throughput pipelines to analyze reports and edit history across petabytes of geographical data, identifying regressions through various signals in internal & user reports to ensure Google Maps data integrity.
- Developed automated validation frameworks to compare pre-edit and post-edit signals, enabling the detection of subtle data regressions and reducing query execution latency by 86%.

Software Engineer Intern

TikTok

Jun 2025 – Nov 2025

Sydney, Australia

- Trust & Safety Engineering in Test, Engagement Safety (Direct Messages)
- Enhanced Python QA frameworks to test Go microservices and drove DM Safety stability & coverage metrics from 6th to 1st out of 14 Trust and Safety subteams, maintaining RPC test stability rates of 99%+ every week.
- Scaled testing infrastructure, developed internal tools, created testing pipelines for the stability of millions of TikTok DM users, cutting cron-job test runtimes by 81%.

Tutor (Teaching Assistant)

UNSW Computer Science & Engineering

Feb 2024 – Present

Sydney, Australia

- COMP2041 (Software Construction [*Linux, Shell, Python*]) Tutor: Overall rating 5.88/6.00 (avg: 5.39/6.00).
- COMP3311 (Database Systems [*SQL, Database Design*]) Tutor: Overall rating 5.67/6.00 (avg: 5.39/6.00).

PROJECTS

q-lite | *C/C++, Operating Systems, High-Performance Computing*

- Made a columnar time-series database in C++20 inspired by kdb+, utilising Structure-of-Arrays (SoA) layout to maximise CPU L1/L2 cache hits during linear scans, with custom static polymorphism to avoid vtable overhead.
- Implemented a high-performance custom arena allocator to ensure cache-local data placement and eliminate heap fragmentation, achieving 12x higher allocation throughput compared to standard `malloc` and `new`.
- Accelerated aggregations using AVX2/SIMD intrinsics and loop unrolling, achieving a 3.5x throughput increase over `std::vector` and `std::unordered_set` through vectorised execution.

SmartTransit | *Python, SQL, AWS, Algorithms, Data Structures, Flask, Stripe, Docker, React Native*

- Created a bus transportation app where bus routes dynamically change according to real-time customer needs.
- Implemented backend infrastructure on AWS using Dockerised Flask services + RDS, with CI/CD deployment pipeline and basic monitoring/logging to support reliable production releases.
- Integrated Stripe for secure payments, and used Google Maps API for live tracking with websockets.

Redis Clone | *C/C++, Socket Programming, Networks, Data Structures, Operating Systems, Redis*

- Built a high-performance Redis clone from scratch, featuring a non-blocking, event-driven architecture using polling for scalable multi-client handling, efficient I/O, and minimal syscall overhead
- Implemented custom protocols for client-server communication, with RESP-style command parsing.
- Explored various client connection models, including multithreading, event-driven polling, and WebSockets to compare scalability, latency, and system resource usage.

TECHNICAL SKILLS

Languages: C/C++, Python, Java, Go, SQL, JS/TS

Systems: Database Systems, Computer Networks, Operating Systems, Compilers, Distributed Systems, HPC

Misc: Data Structures, Algorithms, Object-Oriented Programming, System Design, Git, gRPC, Protobuf, CI/CD, AWS